GATE-2002
BT: BIOTECHNOLOGY

Directions (Q. 1 – 25) : *For each sub-questions given below, four alternatives are provided of which only one is correct. Write the correct answer in the answer book by writing (a), (b), (c) or (d) along with the corresponding sub-question number.* $(25 \times 1 = 25)$

1. The precursor for penicillin-G biosynthesis during fermentation process is
   (a) Phenylacetic acid
   (b) Phenoxyacetic acid
   (c) Acetic acid
   (d) None of these
2. Callus formation from mature tissue explant occurs through
   (a) Dedifferentiation
   (b) Redifferentiation
   (c) Both (a) and (b) of the above
   (d) None of (a) and (b)
3. RNA is very much susceptible to hydrolysis in alkali because
   (a) It contains uracil residues in its structure
   (b) Its 2′ -OH group participate in intramolecular cleavage of phosphodiester backbone
   (c) Cleavage occurs in the glycosylic bonds of purine bases
   (d) Cleavage occurs in the glycosylic bonds of pyrimidine bases
4. The typical coenzyme present in the methanogenes is
   (a) Coenzyme A
   (b) Coenzyme Q
   (c) Coenzyme M
   (d) None of these
5. Large scale clonal propagation practically means raising a population of plantlets from
   (a) A single cell
   (b) A single explant
   (c) Many explants from a single plant
   (d) Many explants from a group of plants
6. T4 Polynucleotide kinase is used for
   (a) Labelling 3′ ends of DNA
   (b) Labelling 5′ ends of DNA
   (c) Creating blunt ends of DNA
   (d) Dephosphorylation of DNA
7. Energy capture efficiency of the aerobic cells using glucose as a substrate is
   (a) 50% (b) 40%
   (c) 30% (d) 20%
8. Plant secondary metabolites production in suspension culture is mainly targeted for
   (a) Obtaining metabolites in aseptic condition
   (b) Enhanced in vitro production of desired metabolite
   (c) Enhanced production of all metabolites
   (d) Obtaining new metabolites
9. A plasmid cloning vector should contains all of the following sequences except
   (a) Origin of replication
   (b) Inducible promoter
   (c) Selectable marker gene
   (d) Multiple cloning sites
10. For scaling up of a bioreactor, the following parameter is assumed to be constant
    (a) Airflow rate
    (b) Diameter of the impeller
    (c) Agitator speed
    (d) Volumetric mass transfer coefficient
11. Stable transformation of plants is reliably obtained by
    (a) Agrobacterium plasmid integration
    (b) Electroporation
    (c) Microinjection
    (d) Silicon carbide whisker
12. In baculovirus expression vector foreign genes are expressed from the promoter of
    (a) Polyhedron gene
    (b) Bacteriophage T7 gene
    (c) *E. coli* lacZ gene
    (d) Yeast phosphoglycerate kinase gene
13. Mechanism of separation of contaminants present in air by fibrous media are
    (a) Interception
    (b) Inertial impaction
    (c) Diffusion
    (d) All of these

14. The length of each boarder sequence in Ti -plasmid is about
    (a) 25 million base pairs
    (b) 200 kilo base pairs
    (c) 25 kilo base pairs
    (d) 25 base pairs
15. Enzyme used in 'cycle' sequencing of DNA is
    (a) T7 DNA polymerase
    (b) T4 DNA polymerase
    (c) Klenow DNA polymerase
    (d) Taq DNA polymerase
16. Ethanol concentration is lowest in
    (a) Wine (b) Beer
    (c) Brandy (d) Rum
17. Plastome is
    (a) A type of plastid
    (b) An organellar genome
    (c) Plasmalemma protein
    (d) None of (a), (b), (c)
18. Recombinant live attenuated vaccine against hepatitis B was prepared from
    (a) Plasma of chronically infected individual
    (b) Recombinant yeast expressing hepatitis B surface antigen
    (c) Recombinant vaccinia virus expressing hepatitis B surface antigen
    (d) Transgenic plants expressing hepatitis B surface antigen
19. The following cross-linking agents may be used for the immobilization of enzymes
    (a) Glutaraldehyde
    (b) Cyanogen bromide
    (c) Thionyl chloride
    (d) All of these
20. Which of the following statements is most appropriate for recombinant antibody production in transgenic plants?
    (a) A very high level expression is always obtained
    (b) Light promote more antibody production
    (c) Such antibodies are free from other antigen of animal origin
    (d) Functional antibody cannot be produced in plants
21. Cells deficient in hypoxanthine guanine phosphoribosyl transferase (HPRT) enzyme rely on
    (a) Synthesis of purine deoxynucleotides by salvage pathway
    (b) Synthesis of purine deoxynucleotides by de novo pathway
    (c) Supply of hypoxanthine in the culture medium
    (d) Supply of thymidine in the culture medium
22. Enhanced axillary branching for multiple shoot productions is promoted by
    (a) 2, 4 – D
    (b) Abscisic acid
    (c) Gibberellic acid
    (d) Benzyl adenine
23. The following culture systems are used for growing large amount of anchorage dependent animal cells except
    (a) Roller bottle
    (b) Airlift fermenter
    (c) Hollow fibre reactor
    (d) Microcarrieres
24. Viral replication within cells is inhibited by
    (a) IL – 4 (b) IL – 1
    (c) IFNα (d) TNFα
25. In large scale fermentation process, air is sterilized by
    (a) Jute fiber
    (b) Membrane
    (c) Cotton fiber
    (d) Glass wool fiber
26. Match the organisms in **Column A** with the product in **Column B**.

Column - A	Column - B
(a) *Thermus aquaticus*	1. Beer
(b) *Acetobacter aceti*	2. Bioinsecticides
(c) *Bacillus thermogenesis*	3. Hind III
(d) *Saccharomyces carlbergensis*	4. Taq 1
(e) *Hemophilus influenza*	5. Vinegar

27. Match the secondary metabolites in **Column A** with their most appropriate chemical characteristics in **Column B**.

Column - A	Column - B
(a) *Diosgenin*	1. Holoside
(b) *Ajmalicine*	2. Pyrrolizidine alkaloid
(c) *Shikonin*	3. Indole alkaloid
(d) *Digoxin*	4. Naphthaquinone nucleus
(e) *Scopolamine*	5. Cardenolide
	6. Saponin
	7. Tropane alkaloid

28. (a) Name two metal ions which play important role in citric acid fermentation. (1)

(b) How is the agitator speed in a fermenter correlated with the power drawn by the agitator? (1)

(c) How does the sulfanilamide kill the bacteria ? (1)

(d) Human insulin gene cloned from a cDNA library into pUC19 could not be expressed. Justify the reason. (1)

(e) Mention the specific role of acetosyringone in *Agrobacterium mediated* plant transformation ? (1)

29. (a) The volume of a chemostat system is 1000/. The feed flow rate to the reactor is 200 *l/h* and the glucose concentration in the feed is 5 g/l. Determine cell and glucose concentration in the effluent of the reactor under steady state conditions. Use the following constants for the cells :

$\mu_{max} = 0.3\ h^{-1}$, $K_s = 0.1$ g/l,

$Y_{x/s} = 0.4$ (g dw cells/g glucose) (3)

(b) Find out the dilution rate which gives maximum biomass productivity.

30. (a) What are the characteristics of 'normal' primary animal cells ? (2)

(b) Name different methods for the separation of different cell types from a mixed population of animal cells ? (1)

(c) What is transgene?

ANSWERS

1. (a)	2. (c)	3. (b)	4. (c)	5. (c)	6. (b)	7. (c)	8. (b)	9. (b)	10. (d)
11. (a)	12. (a)	13. (d)	14. (d)	15. (d)	16. (b)	17. (b)	18. (c)	19. (d)	20. (c)
21. (b)	22. (a)	23. (b)	24. (c)	25. (d)					

Note : Q. No. 26 to 30 are Subjective Questions.

EXPLANATIONS

1. Penicillium chrysogenum utilizes phenylacetic acid as a side chain precursor in penicillin G biosynthesis. During industrial production of penicillin G, phenylacetic acid is fed in small amounts to the medium to avoid toxic side effects. Phenylacetic acid is taken up from the medium and intracellularly coupled to 6-aminopenicillanic acid.

2. Mature tissue explant first undergoes dedifferentiation and then redifferentiation in order to produce a viable callus in cell culture.

3. While DNA contains *deoxyribose*, RNA contains *ribose* (in deoxyribose there is no hydroxyl group attached to the pentose ring in the 2′ position). These hydroxyl groups make RNA less stable than DNA because it is more prone to hydrolysis.

4. Coenzyme M is a coenzyme required for methyl-transfer reactions in the metabolism of methanogens. The coenzyme is an anion with the formula $HSCH_2CH_2SO^-3$. It is named 2-mercaptoethanesulfonate and abbreviated HS–CoM. The cation is unimportant, but the sodium salt is most available. Mercaptoethanesulfonate contains both a thiol, which is the main site of reactivity, and a sulfonate group, which confers solubility in aqueous media.

5. Clonal propagation is the asexual propagation of the plantlets. It needs less time and a single plant to generate a pure verity of the plant.

6. T4 Polynucleotide Kinase catalyzes the transfer and exchange of P_i from the γ position of ATP to the 5′-hydroxyl terminus of double- and single-stranded DNA and RNA, as well as nucleoside 3′-monophosphates. This modified version exhibits full kinase activity with no 3′ phosphatase activity.

Highlights of T4 Polynucleotide Kinase:

(1) 5′ phosphorylation of DNA/RNA for subsequent ligation

(2) End-labeling of DNA or RNA

(3) 5′ phosphorylation of 3′ phosphorylated mononucleotides to generate a substrate (pNp) that can be added to the 3′ end of DNA or RNA by ligase activity

(4) 5′ end labeling of 3′ phosphorylated oligos

7. Complete aerobic metabolism of glucose produces water and carbon dioxide as products.

$$C_6H_{12}O_6 + 6\,O_2 \rightarrow 6\,CO_2 + 6\,H_2O$$

Energy is released in this process. The overall "G of glucose breakdown is -720 kcal/mole. Normally about 32% of the energy is released is captured through the formation of ATP the remainder is released as heat.

8. The secondary metabolism can be regulated to maximize the production of target compounds.

9. plasmid vector should not contain inducible promoter at the time of cloning. As it may induce secondary metabolic pathway activation.

10. Volumetric mass transfer coefficient should be constant for scaling up process in fermentation.

11. Agrobacterium plasmid contains Ti-gene, Nop gene and Ori C which are crucial for stable gene transfer procedure.

12. Baculoviruses are insect pathogenic and insect specific viruses belonging to the group of Nuclear polyhedrosis viruses. Cloning vectors exploit the extraordinary strength of the polyhedrin gene promoter.

13. The fibrous filter media are known to capture the airborne particles of different sizes due to Brownian diffusion, direct interception, inertial impaction, gravity settling, and particle sieving mechanisms.

14. The border sequence of Ti plasmid is known to be around 25bp long which is sufficient for cleavage and overdrive sequences.

15. Taq polymerase is necessary for DNA cycle sequencing.

16. Amount of ethanol in brandy: 35 to 60%, rum: 40%, beer: 4 to 6%.

17. The *plastome* is the genetic material that is found in plastids in plant cells (for example in the chloroplast). It composes part of the entire genome of photosynthetic organisms.

18. A vaccine vector, or carrier, is weekend virus or bacterium into which harmless genetic material from another disease causing organism can be inserted. The vaccine virus, the virus that causes cowpox, is now used to make recombinant vector vaccines. Vaccinia is relatively large and has ample room to accept additional genetic fragments.

19. Commonly used methods for the covalent immobilisation of enzymes includes activation of Sepharose by cyanogen bromide. Conditions are chosen to minimise the formation of the inert carbamate. Glutaraldehyde is used to cross-link enzymes or link them to supports. It usually consists of an equilibrium mixture of monomer and oligomers. The product of the condensation of enzyme and glutaraldehyde may be stabilised against dissociation by reduction with sodium borohydride.

20. Its cross reactivity free in plant as such antibodies can be susceptible for cross reactivity in animal system.

21. The antibiotic aminopterin inhibits the de novo nucleotide synthesis pathway. However, normal cells survive in this medium as they are able to use the salvage pathway for nucleic acid synthesis. But if the cells are unable to produce the enzyme Hypoxanthine-Guanine Phospho Ribosyl Transferase (HGPRT), they are unable to utilize the salvage pathway and therefore die in aminopterin containing medium.

22. 2,4-D, being a phenoxy auxin, promoted quick callus formation from the induced roots and also at the base of the shoots.

23. Airlift fermenter is a fermenter in which circulation of the culture medium and aeration is achieved by injection of air into some lower part of the fermenter. Usually not suitable for animal cell production.

24. Interferons (IFNs) are proteins made and released by host cells in response to the presence of pathogens such as viruses, bacteria, parasites ortumor cells. They allow for communication between cells to trigger the protective defenses of the immune system that eradicate pathogens or tumors.

25. To prevent contamination of either the fermentation by air-borne microorganisms or the environment by aerosols generated within the fermenter, both air input and air exhaust ports have air filters such as glass fibre filters, mineral fibers filters.